



TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Tiny FPGA development board

The C0-microSD is a compact development board from Signaloid featuring an iCE40 FPGA. Its distinctive feature is its ability to fit into a standard microSD



card socket, which also serves as its programming interface. The compact size of this board is

comparable to a microSD card, with the LEDs positioned at the top edge for visibility when inserted into a socket. What sets this development board apart is its microSD interface. When plugged into a host computer, it is recognised as a file-based device. This allows for programming the board directly through the host's operating system without needing an external programmer. It is designed for engineers, developers, and researchers involved in FPGA development, edge computing, uncertainty analysis, and embedded systems. Its compact size and unique programming model make it suitable for applications in wearables, IoT devices, and predictive modelling.

Signaloid

<https://signaloid.com>

The 'world's fastest' CPU

Intel has launched the Intel Core 14th Gen i9-14900KS processors, claiming the title of the world's fastest desktop processor with a max turbo frequency of up to 6.2 gigahertz (GHz). Designed for desktop enthusiasts seeking maximum



power, this processor offers a premium gaming and creative experience. It enables up to 15% better performance generation-over-generation for gamers,

thanks to its high speeds and Intel's application performance optimisation (APO) feature. Content creators will also benefit from its performance, with up to 73% improvement in compute-intensive workflows like 3D production multitasking compared to competitors. It is also suitable for professional workstations in fields such as engineering and scientific research, where high performance is critical.

Intel

<https://www.intel.com>

The 'world's most powerful chip' for AI

NVIDIA has launched the Blackwell platform, marking a new era in computing by enabling organisations



to operate real-time generative AI on large language models with trillion parameters at up to 25x less cost and energy consumption compared to its predecessor. The Black-

well GPU architecture incorporates six groundbreaking technologies for accelerated computing, poised to drive advancements in various fields, including data processing, engineering simulation, and generative AI. It is aimed at a diverse range of users, including cloud service providers. It can also be used by data centres, enterprises, AI developers, and server manufacturers.

NVIDIA

<https://www.nvidia.com/en-in>

Multi-band LoRa radio module

Murata has introduced the Type 2GT module, a new low-power, multi-band LoRa module for IoT devices. This module stands out for its ability

to meet global certification standards, making it easier to use in different regions and industries. The module is small, at 9.98mm x 8.70mm x 1.74mm, and is built on a PCB inside a metal



case. For designers, this means simpler PCB designs, faster time to market, and lower devel-

opment costs. The module is designed for developers and innovators creating IoT devices that need low-power, long-range wireless connectivity. It is suitable for applications in smart agriculture, industrial and environmental sensing, building and home automation, asset tracking, and utilities metering. The module's support for multiple frequency bands and global certification standards makes it versatile and easy to deploy in various regions and industries.

Murata

<https://www.murata.com>

Enhanced storage for AI PCs

The YRS820, based on the RISC-V architecture, features a 4-channel PCIe 5.0 interface, 8 NAND flash memory channels, supports NVMe 2.0 protocol, and offers an interface transmission rate



of 2667MT/s. It supports 3D TLC/QLC and up to 8TB capacity. Its performance includes 14GB/s

sequential read, 12GB/s sequential write, and up to 2000k IOPs random read and 1500k IOPs random write. The SSD controller is for consumer markets, especially AI PCs. It is for gamers, professionals in data-intensive applications like content creation and analytics, and users of AI and machine learning appli-